Learning Lifted Symbolic Planning Domains from Image Sequences

VISA: Visual-to-Symbolic Action Learning

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Motivation: Enabling Generalizable Visual Planning

Symbolic planners require action models

- But these are usually **handcrafted**.
- **Not naturally grounded** in raw *visual observations*.

Prior visual model-learning approaches assume

- A **fixed number of objects**, *limiting generalization*.
- Training traces contain **fully grounded STRIPS actions** with *all arguments specified*.

Objective: Learn lifted domains directly from images, enabling generalization to *larger* planning instances.

Why Is This Difficult?

1. Grounding

- Images contain pixels, not symbolic predicates.
- Object identities and relations are not given.

2. Lifting

- Object counts change across planning instances.
- Fixed propositional encodings do not transfer.

3. Partial Actions

- Traces may omit action arguments.
- They are not full STRIPS action traces.

The learned representation must be *grounded*, *relational*, and *lifted*.

Previous Work: Three Pieces of the Puzzle

LatPlan

✓ Learns latent, propositional action models from state image pairs

✓ Does not rely on a known symbolic vocabulary

× Learned model is uninterpretable

× Latent propositional representation is tied to a fixed object set

ROSAME

✓ Learns lifted, human-readable action models from visual traces

✓ Computes probabilistic preconditions and effects for action schemas

× Executed actions are fully observed

× Requires full STRIPS action traces

ROSAME v.2

✓ Jointly learns to predict states, executed actions, and a lifted action model

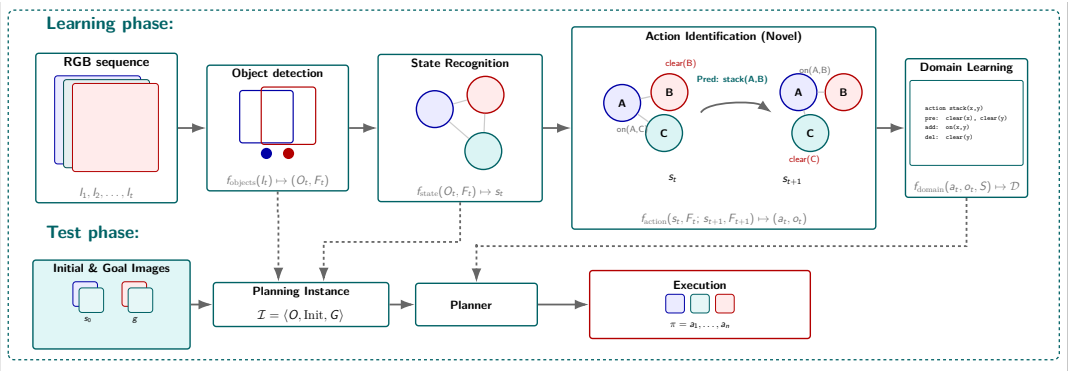
✓ Removes the need for observed action labels

× Relies on costly MILP-based correction during training

× Hard to scale to longer traces

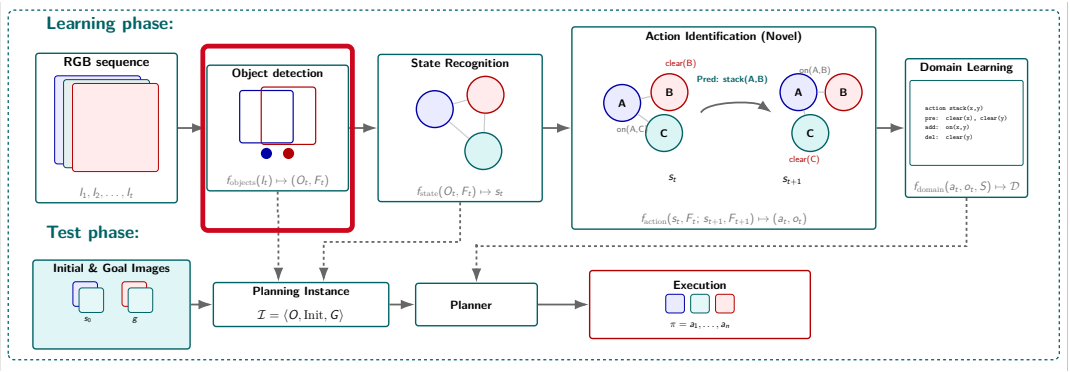
VISA differs by grounding images into *dynamic relational scene graphs* and learning lifted domains from *partially* specified visual *action traces*. Asai et al., JAIR'22 Xi et al., ICAPS'24 Xi et al., ICAPS'26

VISA at a Glance



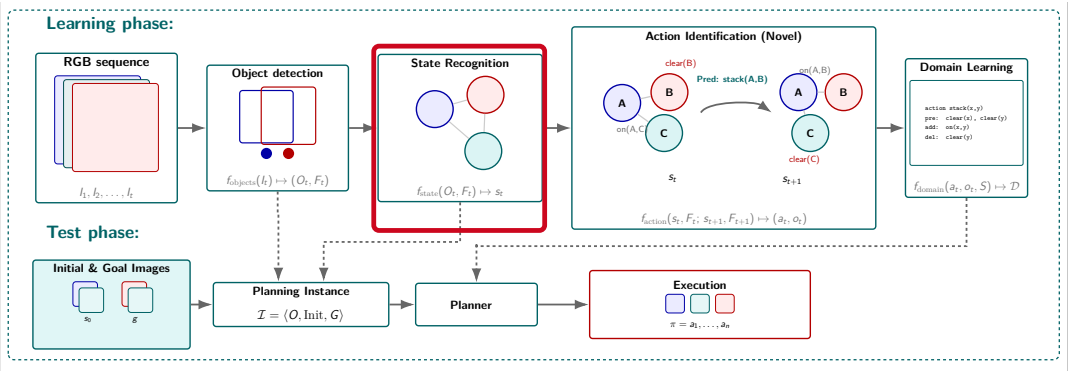
One architecture, two phases: learn a lifted domain from visual traces, then reuse it to plan from new images.

VISA at a Glance



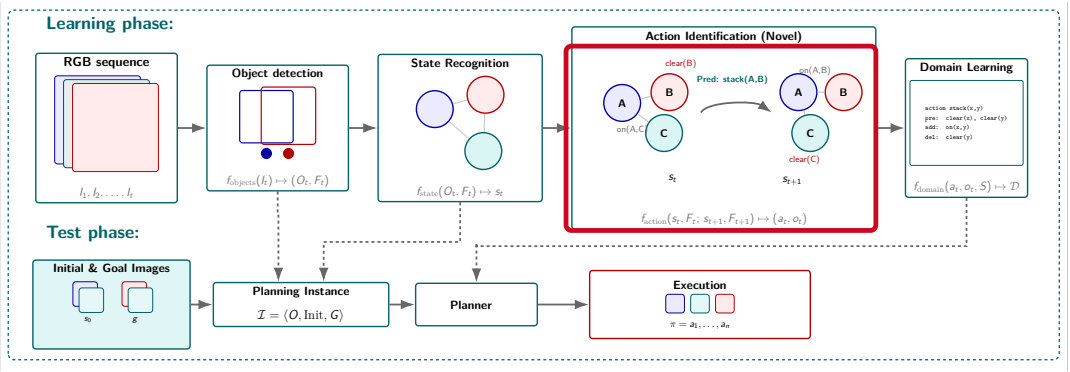
1. **Object detection** turns every frame into a variable-size set of object instances and visual features.

VISA at a Glance



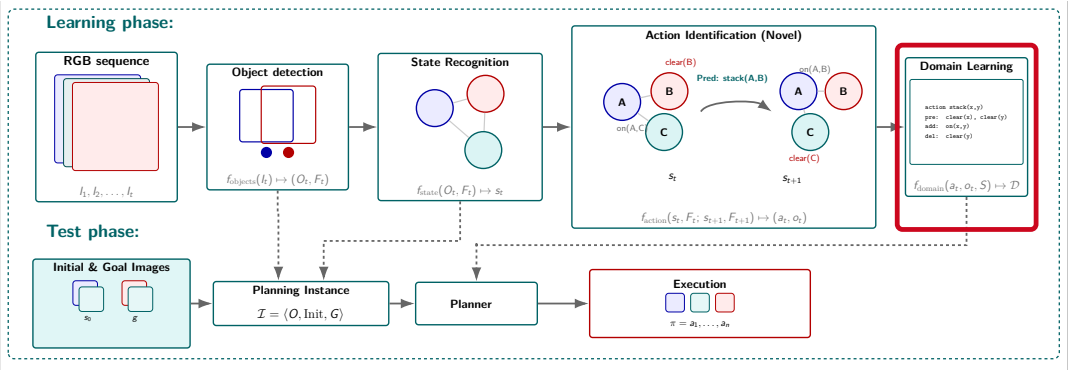
2. **State recognition** converts detected objects into a relational scene graph interpreted as a symbolic state.

VISA at a Glance



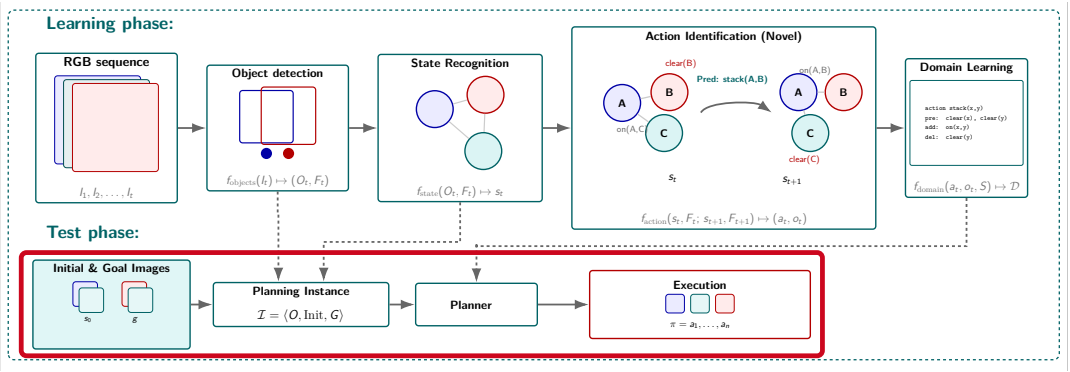
3. Action identification explains each graph transition with an action type and its object arguments.

VISA at a Glance



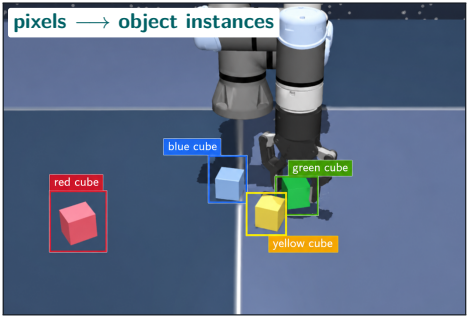
4. Domain learning lifts state–action traces into reusable STRIPS+ action schemas.

VISA at a Glance



5. Planning combines a learned domain with new initial and goal images to produce a plan.

1. Object Detection: Pixels Become Objects

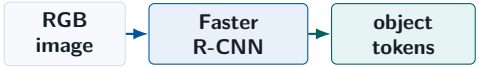


$$I_t \longmapsto O_t = \{(\hat{c}_j, \mathbf{b}_j, \mathbf{f}_j)\}_{j=1}^{|O_t|}$$

class • box • feature vector

We use Faster R-CNN

fine-tuned for each domain

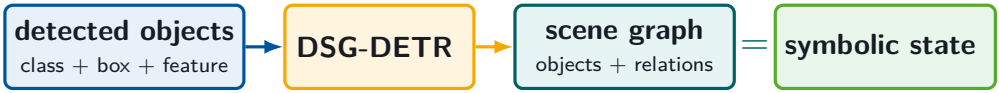


- CNN backbone extracts image features.
- RPN proposes candidate objects.
- Detection head predicts class and box.
- RoI feature is retained as $\mathbf{f}_j$.

Keep $\mathbf{f}_j$: each object's learned feature is passed to **relation prediction** and **action prediction**.

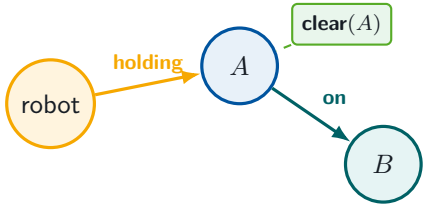
Ren et al., NeurIPS 2015

2. State Recognition: Objects Become a Symbolic State



What DSG-DETR does

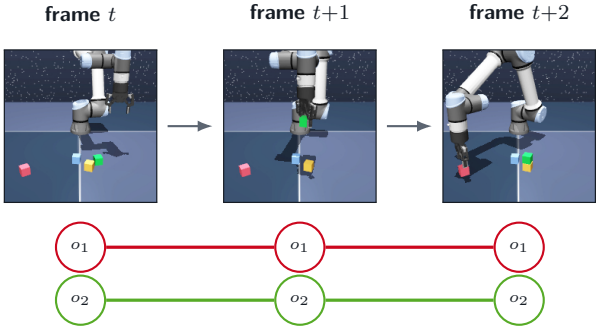
- Builds features for each **ordered object pair**.
- A spatial transformer reasons over relations within the frame.
- A classifier predicts **unary and binary predicates**.



A scene graph is interpreted directly as the symbolic planning state.

Feng et al., WACV'23 (DSG-DETR)

2. State Recognition: Why the Graph Is Dynamic



Temporal consistency

- Associate detections across frames.
- Temporal attention refines identities and relations.

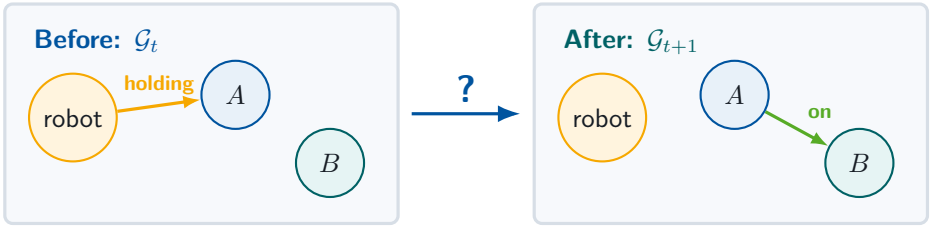
$$\mathcal{G}^{\text{dyn}} = \{\mathcal{G}_1, \ldots, \mathcal{G}_T\}$$
$$\mathcal{G}_t = (\mathcal{V}_t, \mathcal{E}_t)$$

$\mathcal{V}_t$: objects $\quad$ $\mathcal{E}_t$: relations

Planning operates on relational scene graphs, not pixels.

Feng et al., WACV'23

3. Action Identification: What Caused This Transition?



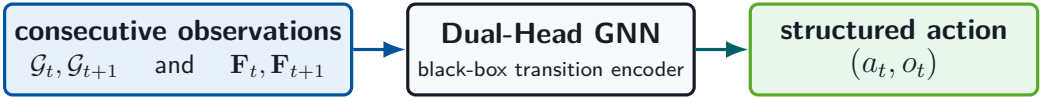
$\Longrightarrow$

$-$ **holding**$(robot, A)$
$+$ **on**(A, B)

structured action
place(A, B)

Action identification is transition explanation: infer the action that best accounts for the changed relations.

3. Action Identification: Dual-Head Structured Prediction



Head 1: action type classifier

$$a_t = \textit{place}$$

e.g. place, pickup, right

Head 2: argument selector

$$o_t = \{A, B\}$$

selects participating objects

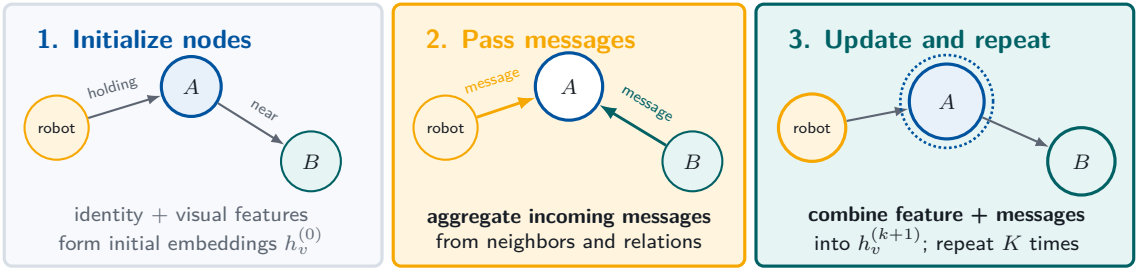
Each graph transition becomes an action observation: type + object arguments.

These observations are the input to domain learning.

Yan et al., AAAI'18 (ST-GCN)

3. Action Identification: How the Network Reasons

A graph neural network lets each object learn from its relational neighborhood.



After K rounds, each node represents itself *in context* – not in isolation.

Gilmer et al., ICML'17 (message-passing neural networks)

4. Domain Learning: How SYNTH Recovers Schemas

Visual traces may be incomplete

Only directly observed action arguments are given.

$$\textit{move\_right}()$$

instead of $\textit{move}(tile, from, to)$

SYNTH searches for a lifted explanation

It finds schemas whose preconditions and effects consistently explain the observed transitions.

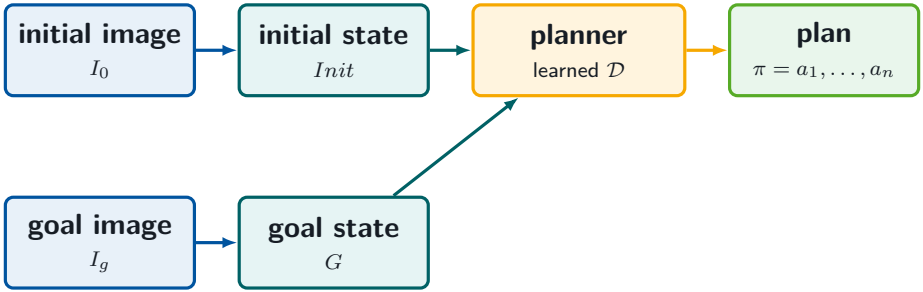
$$\textit{move}(x, y, z)$$

missing arguments bound by state relations

Key idea: STRIPS+ keeps traces partial; SYNTH recovers the hidden structure needed for lifted schemas.

Jansen et al., KR'25 (SYNTH / STRIPS+)

Planning from Images with the Learned Domain



Training

Learn domain $\mathcal{D}$ once from visual demonstrations translated into state–action traces.

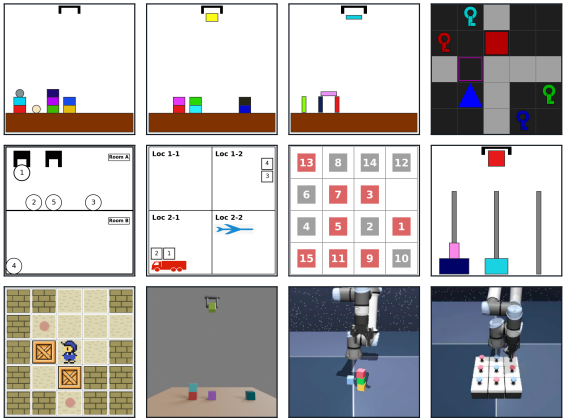
Deployment

Translate new initial/goal images into states and solve with a standard symbolic planner.

Experimental Datasets

Automatic dataset generation

- Use **PDDLGym** planning domains.
- Randomly sample an initial and goal state.
- Solve the planning problem with a classical planner.
- Execute the plan to generate image sequences.
- Automatically record all symbolic annotations.



PDDLGym planning domains (left) and OGBench robotic manipulation environments (right).

Silver & Chitnis, PRL@ICAPS'20 (PDDLGym) Park et al., ICLR'25 (OGBench)

Main Results

Type	Domain	N_O	N_R	N_A	Objects	Frames	Seq.	mAP	F1	ac_{type}	ac_{args}	VAL
Synthetic	BallHiding	4	8	6	5–12	8.6k	4–19	1.00	.99	100	99.82	1.00
Synthetic	Blocksworld	3	6	4	5–12	8.3k	2–23	1.00	1.00	100	99.35	1.00
Synthetic	BW-Realistic	3	6	4	5–12	8.3k	2–23	.99	.99	100	93.94	1.00
Synthetic	Dominoes	3	13	13	5–7	8.3k	2–16	1.00	1.00	100	100	1.00
Synthetic	Grid	1	16	20	25	5.2k	2–10	1.00	.98	99.65	99.84	1.00
Synthetic	Gripper	3	6	4	9–10	8.1k	2–10	1.00	1.00	100	100	1.00
Synthetic	Logistics	4	3	6	10–11	8.4k	2–17	1.00	1.00	99.73	91.18	1.00
Synthetic	SlidingTiles	3	5	4	18–32	8.8k	2–10	1.00	1.00	100	N/A	1.00
Synthetic	Sokoban-undir.	1	4	2	25	6.8k	4–12	1.00	1.00	100	100	1.00
Synthetic	Hanoi	3	6	2	6–9	10.7k	2–31	1.00	.99	100	100	1.00
Interpol.	Blocksworld	3	6	7	5–12	26.1k	2–31	1.00	1.00	99.95	99.91	N/A
Interpol.	BW-Realistic	3	6	7	5–12	26.1k	2–31	1.00	.99	99.24	96.42	N/A
OGBench	Permuting Cubes	3	4	2	2–3	2.3k	2–6	.88	.73	83.80	70.52	.67
OGBench	Stacking Cubes	3	6	4	2–3	3.5k	2–8	.68	.52	58.18	43.54	OR
OGBench	Puzzle	2	4	2	9	6.2k	2–8	.53	.37	21.30	13.72	OR

OR = object recognition failure.

Generalization (OOD) Results

Domain	O_{train}	O_{test}	Grid$_{train}$	Grid$_{test}$	F1	ac_{type}	ac_{args}	VAL
BallHiding	5–8	10–11	N/A	N/A	**.89**	**99.76**	**96.25**	**.80**
Blocksworld	4–8	9–11	N/A	N/A	**1.00**	**99.57**	**100**	**1.00**
BW-Realistic	4–8	9–11	N/A	N/A	**.99**	**96.22**	**85.90**	**1.00**
Grid	25	36	5×5	6×6	**.94**	**89.51**	**92.68**	**.85**
Gripper	7–8	9–10	N/A	N/A	**1.00**	**100**	**100**	**1.00**
Logistics	8–10	11–12	N/A	N/A	**1.00**	**99.23**	**95.51**	**1.00**
SlidingTiles	18	32	3×3	4×4	**.84**	**97.41**	N/A	**.80**
Sokoban-undir.	25	36	5×5	6×6	**.83**	**79.45**	**38.08**	**SR**
Sokoban-undir. goals	25/2	25/3	5×5	5×5	**.99**	**99.90**	**99.90**	**1.00**
Hanoi	6–8	9	N/A	N/A	**1.00**	**100**	**100**	**1.00**
Blocksworld interpol.	4–8	9–11	N/A	N/A	**.98**	**99.88**	**97.27**	N/A
BW-Realistic interpol.	4–8	9–11	N/A	N/A	**.99**	**98.77**	**95.64**	N/A

SR = state representation not compatible with the target planning formalism.

Limitations & Future Work

- **Perception bottleneck:** Low-resolution or cluttered images cause object-detection errors, which downstream symbolic modules cannot fully recover from.
- **Supervision bottleneck:** VISA requires supervised annotations for learning visual relations and action identification, although domain learning can use partially observed states and does not require exhaustive STRIPS labeling.
- **Scalability bottleneck:** State and action modules use attention over object pairs: n objects yield $O(n^2)$ pair embeddings, and attention over these pairs can require up to $O(n^4)$ memory for trajectories.
- STRIPS-**alignment:** Some learned predicates induce transitions that cannot be expressed with fixed add/delete action schemas.

Park et al., ICLR'25 (OGBench)

Takeaways

1. Ground images as object-relation graphs.

2. Predict actions as structured graph transitions.

3. Learn lifted schemas and keep classical planners in the loop.

VISA is a simple, direct approach for learning action models from variable-object image traces.

Thank you.

Appendix: Module Mappings

$$
\begin{aligned}
f_{\text{objects}} &: I_t \mapsto (O_t, \mathbf{F}_t) \\
f_{\text{state}} &: (O_t, \mathbf{F}_t) \mapsto s_t = \mathcal{G}_t \\
f_{\text{action}} &: (s_t, \mathbf{F}_t, s_{t+1}, \mathbf{F}_{t+1}) \mapsto (a_t, o_t) \\
f_{\text{domain}} &: (\{(a_t, o_t)\}, \{s_t\}) \mapsto \mathcal{D}
\end{aligned}
$$

At test time, initial and goal images are translated into $\langle O, Init, G \rangle$, paired with the learned domain $\mathcal{D}$, and solved by a symbolic planner.

Appendix: Talk Timing

One-sentence answer

VISA learns a lifted planning domain from visual traces by converting images into relational scene graphs, predicting structured actions between graph states, and learning STRIPS+ schemas from the resulting traces.